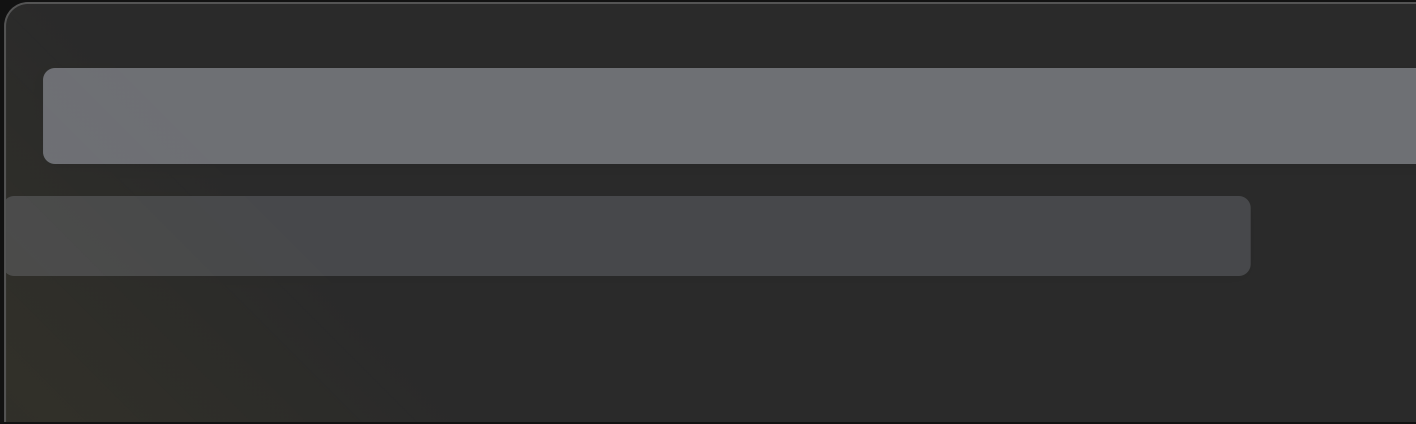
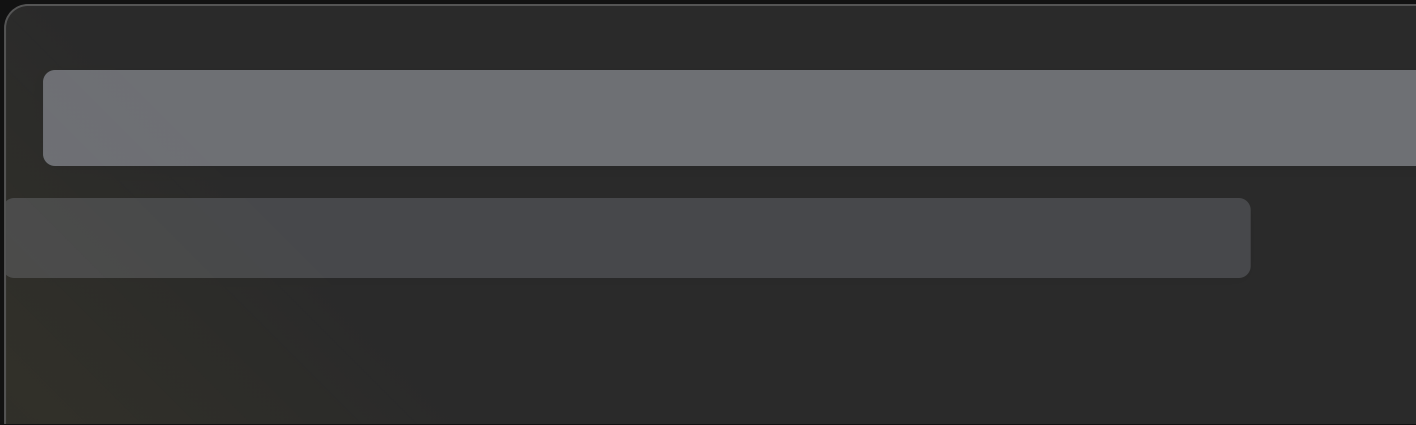




# At first the web was flat



# Today, the web can overlap



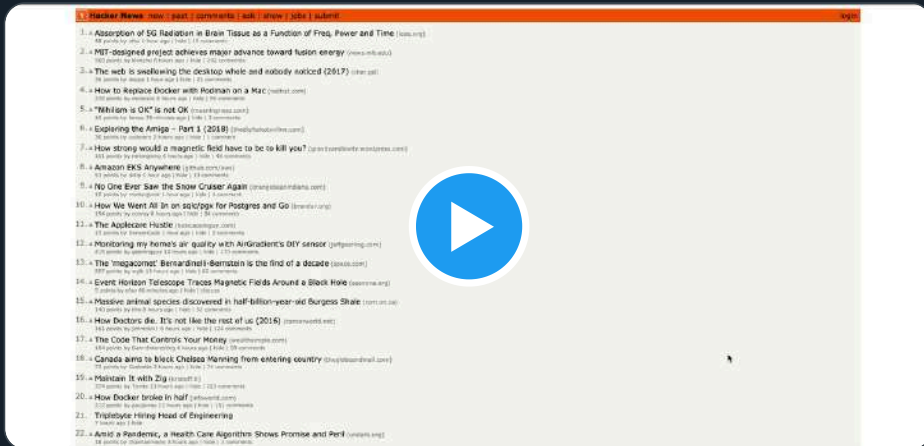


Andrej Karpathy

@karpathy · Follow



## Browsing the web, 2021



6:47 AM · Sep 9, 2021



13.1K



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Bekijken op YouTube

# Tim Damen

👤 Head of Front-end @ focusing.io

🧩 Ecosystem Team Member  @ Nuxt

🌸 Husband and proud father of a daughter

🏔️ Passionate about sports and outdoor adventures





# **We'll focus on the *What*, *Why* and *How* of content that overlaps content**

With the `<dialog>` element and the `[popover]` attribute

# <dialog>

HTML element - a native dialog element that can be used to create a modal dialog or a non-modal dialog.

- Supported by all major browsers
  - 2014: Chrome v37
  - 2022: Firefox 98
  - 2022: Safari 15.4
- Good accessibility support
  - 2014-2021: "The a11y community says nope"
  - 2022: "Universal browser support... but still wait"
  - 2023: "Use the dialog element (reasonably) by Scott ohara"

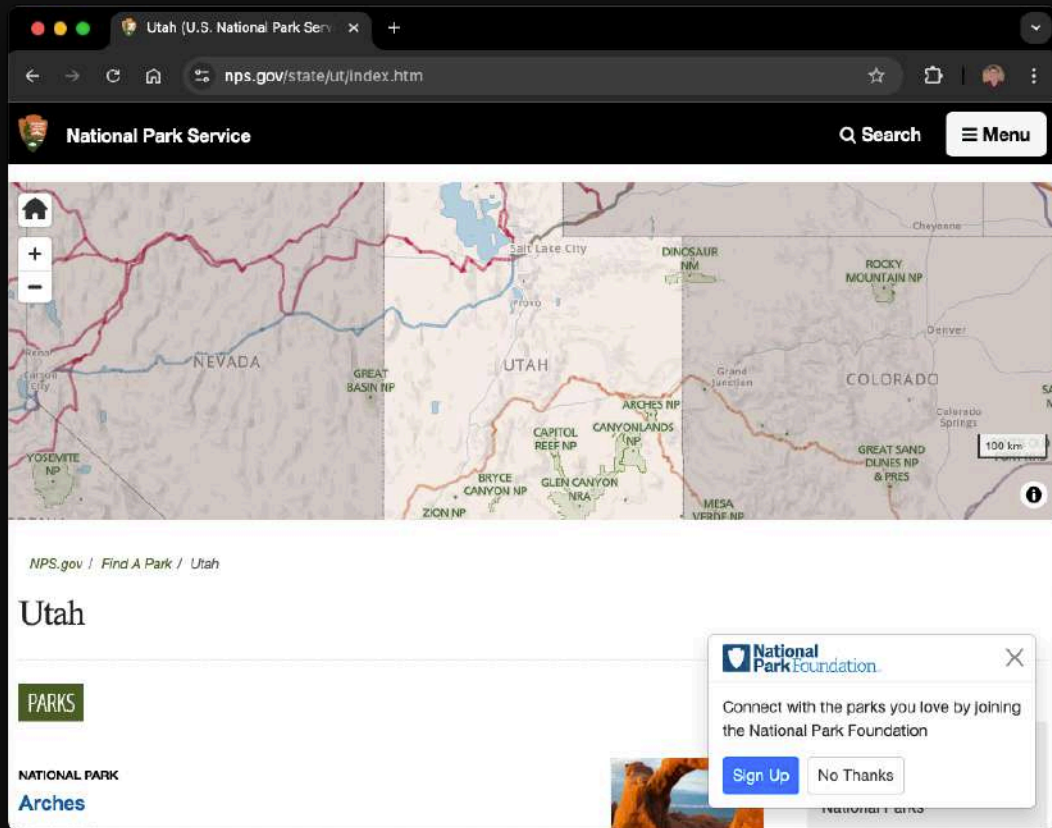


# [popover]

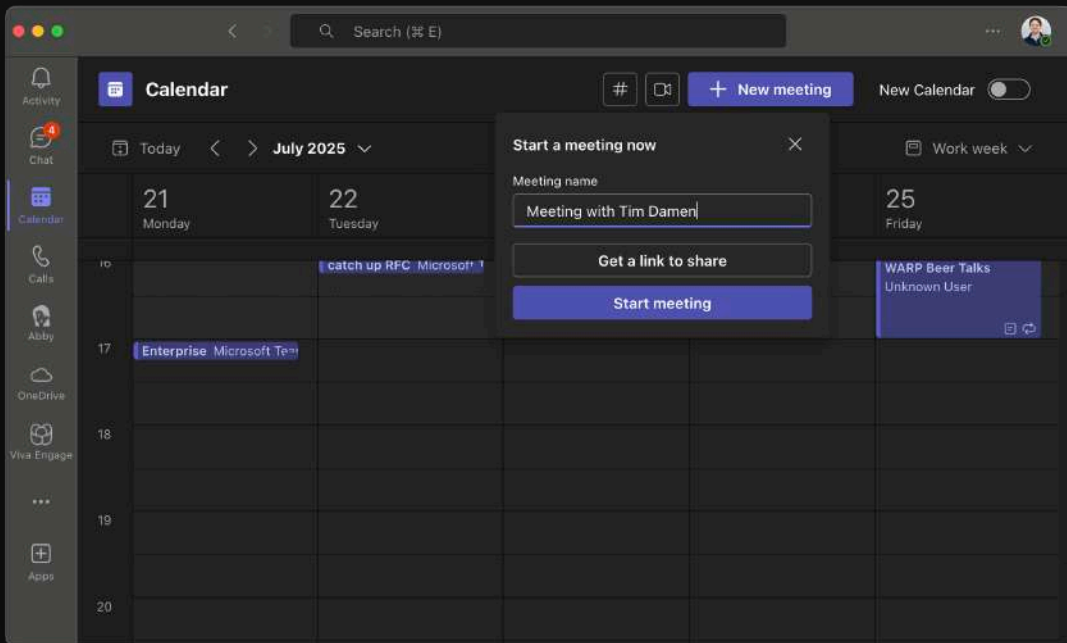
HTML attribute - API in HTML that allows you to create popovers, tooltips, and other overlay content.

- Supported by all major browsers
  - May 2023: Chrome v114
  - Sep 2023: Safari v17
  - Apr 2024: Firefox v125
- Accessibility was baked in from the start
- Proposed by the Open UI group

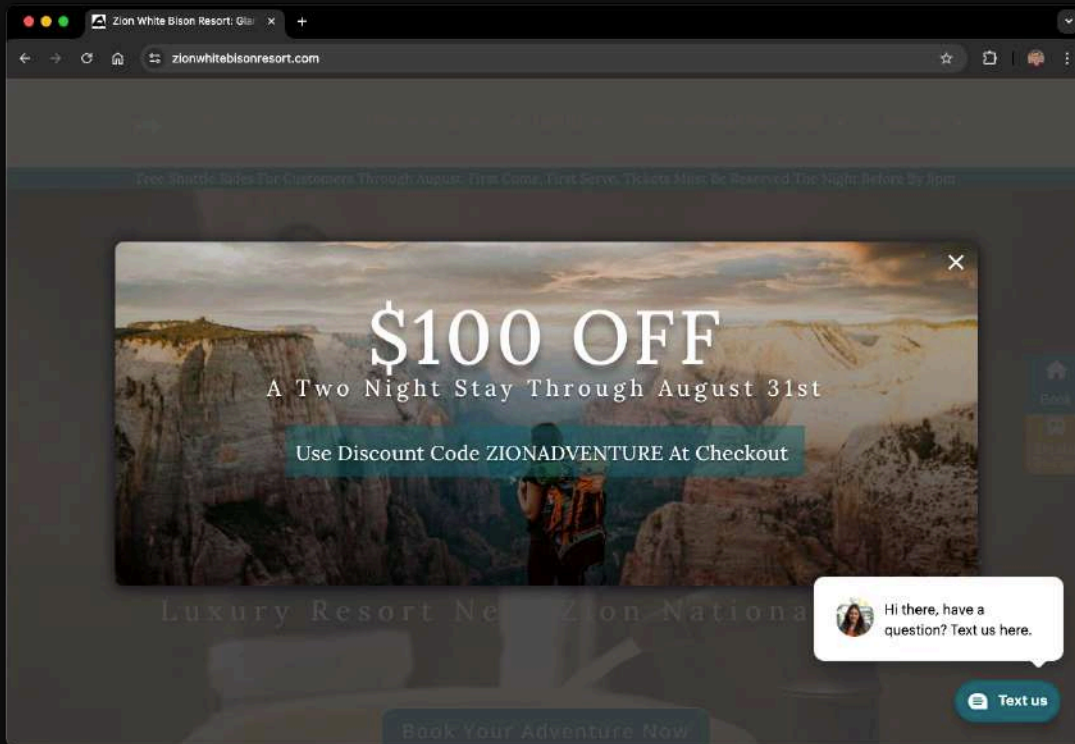
# Non-modal dialog



# Popover



# Modal + Popover



# Before `z-index`: the `<layer>` element

- **Netscape 4 (1997)** introduced the `<layer>` element
  - Allowed absolute positioning and stacking of content
  - Each `<layer>` was a separate document object in JavaScript
  - **Only worked in Netscape** — `if (document.layers)` was true in Netscape 4, but false in all other browsers
- Meanwhile, **CSS2 (1998)** standardised `z-index` for stacking
  - Supported across browsers, no vendor lock-in
  - The `<layer>` element quietly disappeared
- Today `<layer>` is a relic of the Browser Wars — proof that **open standards outlast**

# With `z-index` you can stack elements

```
z-index: auto;
```

```
z-index: 1;
```

```
z-index: 3;
```

```
z-index: 5;
```

```
z-index: 7;
```

```
z-index: 6
```

```
z-index: 4
```

```
z-index: 2
```

Change my z-index

```
z-index: auto
```

```
z-index: auto
```

```
z-index: auto
```

# What is the maximum `z-index` value?

```
z-index: 2147483647; /* 2^31 - 1 */
```

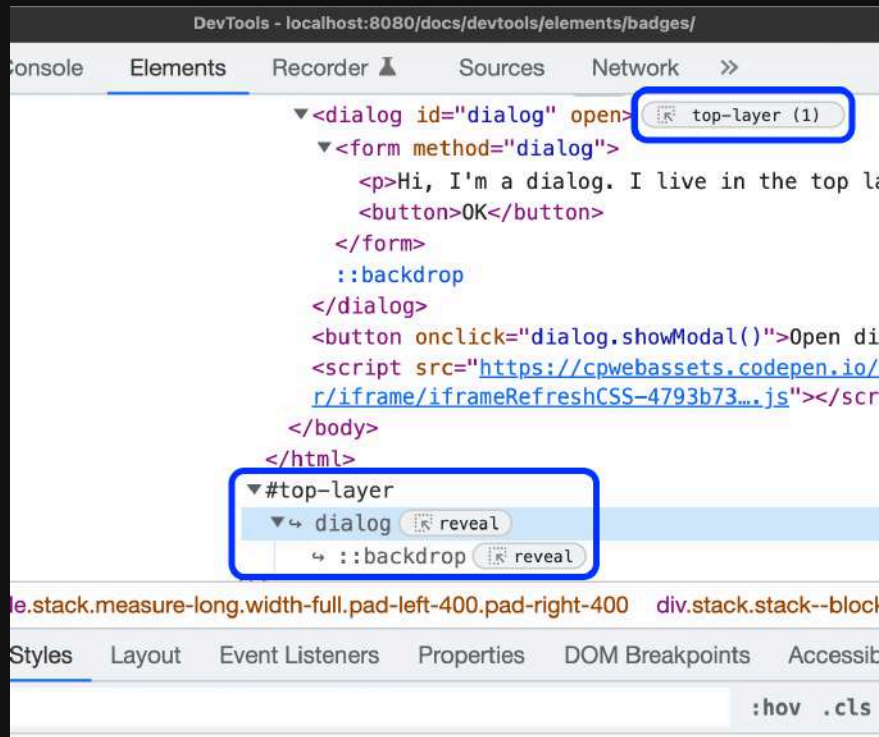


Speak the Number

2,147,483,647

2147483647 , the largest number that can be stored in a 32-bit signed integer 🧐

The `#top-layer` is above everything else, it has its own layer above the main document





## #top-layer

On top of z-index stacking

<dialog>  
dialog

requestFullscreen()  
fullscreen

[popover]  
popover

## z-index

z-index: 1 → 2147483647

Dropdown

z: 10

Tooltip

z: 100

Notification

z: 9999

Overlay

z: 1000

## document

z-index: auto

Content

main

Header

header

Navigation

nav

Sidebar

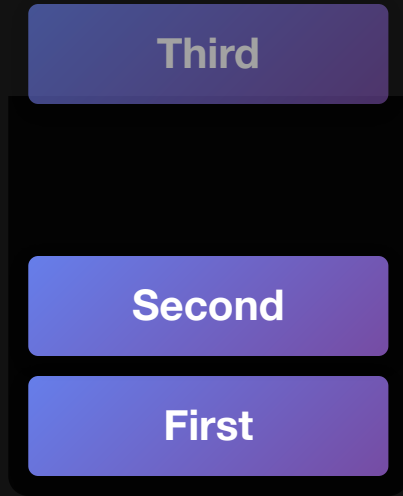
aside

Footer

footer

# FIFO stacking

First In, Last Out



**Push Order:** First → Second → Third

**Pop Order:** Third → Second → First

The first item added is the last to be removed!

<dialog> **vs** [popover]

**What a `<dialog>` is**

**An additional window  
that (temporarily)  
overlaps the main  
content of a web page**

# Dialogs

Typically contain:

- Information
- Task
- Action

# <dialog>

## show() vs showModal()

```
// Opens the dialog as a non-modal  
Element.show();
```

```
// Opens the dialog as a modal  
Element.showModal();
```

```
<!-- dialog element -->  
<dialog>  
  ...  
</dialog>
```

- `role="dialog"` is added automatically
- Option to show as modal or non-modal
- `Esc` to close when opened as modal
- Focus is trapped when opened as modal

# modal vs non-modal

## HTML Dialog Element Demo

Survey dialog

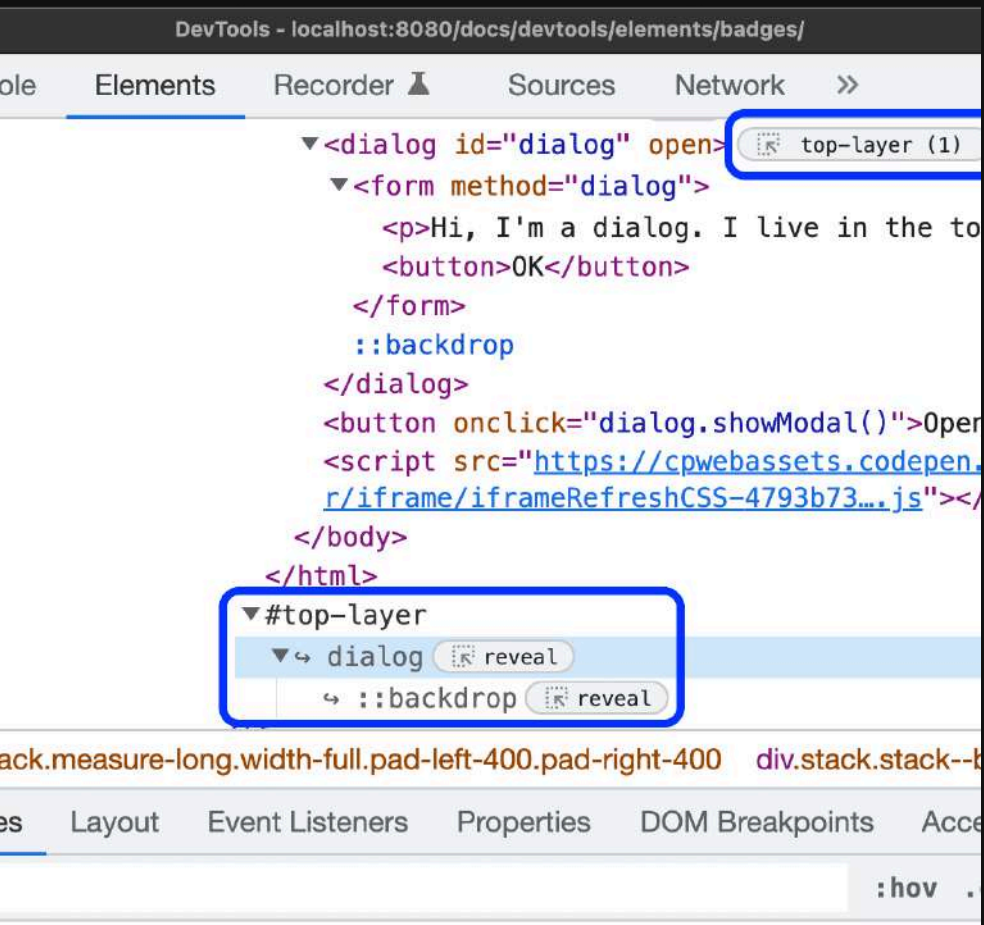
Session extend





## Keyboard focus trap

Sometimes you want to prevent keyboard focus from leaving a component, like a modal dialog.



# backdrop

Some component have a backdrop.

**Toplayer** elements have a built-in styleable pseudo-element

`::backdrop`

`<dialog>` **vs** `[popover]`

**What a [popover] is**

**A floating overlay that provides information or actions when triggered, appearing above the main content without blocking the user's workflow.**

**"a popover 'pops'" - Tim Damen**

# [popover]

**HTML attribute that can be applied to any element, it adds a set of behaviors to the element**

```
<div popover>...</div>
```

```
<dialog popover>...</dialog>
```

Popover does not add a `role`. It adopts the role of the element it is applied to.

Depending on the popover type (default is `popover="auto"`), it will add:

- light dismiss
- keyboard behavior (`Esc` to close)
- added to toplayer
- Closes other popovers when opened
  - (Usually) displayed one at a time
  - (Usually) displayed on hover or click

# [popover] implementation

Requires no JavaScript, just HTML

```
<button>
  Toggle popover
</button>

<div>
  ...
</div>
```

# [popover] implementation

Requires no JavaScript, just HTML

```
<button popovertarget="x" popovertargetaction="show">  
  Open popover  
</button>  
  
<div popover id="x">  
  ...  
</div>
```

Open popover

# [popover] implementation

Requires no JavaScript, just HTML

```
<button popovertarget="x" popovertargetaction="show">
  Open popover
</button>

<!-- Or open with Javascript -->
<button onclick="document.getElementById('x').showPopover()">
  element.showPopover();
</button>

<div popover id="x">
  ...
</div>
```

Open popover

element.showPopover();



localhost:3031/32

Relaunch to update

Elements

31 24 57

style="color: black; padding: 2px; background-color: rgb(240, 240, 240);">element.showPopover(); </button>

...><div data-v-e16fe63a popover id="x">...</div> == \$0

dev-page.slidev-page-32

div.slidev-layout.default

div#x

Styles

Computed

Layout

Event Listeners

Filter

:hov .cls

border-color: var(--un-default-border-color, #e5e7eb);

[popover]:popover-open:not(dialog) user agent stylesheet

overlay: auto !important;

[popover] { user agent stylesheet

position: fixed;

width: fit-content;

height: fit-content;

color: canvastext;

background-color: canvas;

inset: 0px;

margin: auto;

border-width: initial;

border-style: solid;

border-color: initial;

border-image: initial;

padding: 0.25em;

overflow: auto;

div { user agent stylesheet

[popover] implementation

Requires no JavaScript, just HTML

<button popovertarget="x" popovertargetaction="show">Open popover</button>

<!-- 01 Lorem Ipsum is simply dummy text of the printing and typesetting industry. --><button onclick="document.getElementById('x').showPopover()">element.showPopover();</button>

<div popover id="x">...</div>

Open popover element.showPopover();

div#x 596.08 x 35.2

32 / 74

# `[popover]` still has some positioning flaws

**But there is hope! You have got two options:**

- Use a JS library for positioning like `floating-ui`
- Use `CSS anchor()`

`CSS anchor()` is not supported by all browsers yet 😞



# "auto" and "auto"

popover="auto"



popover="auto"



# "auto" and "manual"

popover="auto"



popover="manual"



Show Success

Show Warning

Show Error

# <dialog> vs [popover] vs modals

| <code>&lt;dialog&gt;</code> s<br><code>&lt;dialog&gt;</code> with <code>show()</code> | <code>[popover]</code> s | modals<br><code>&lt;dialog&gt;</code> with <code>showModal()</code> |
|---------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------|
| Explicit dismiss                                                                      | Light dismiss            | Explicit dismiss                                                    |
| z-index                                                                               | #top-layer               | #top-layer                                                          |
|      | ··backdrop ?!            | ··backdrop                                                          |

# The Evolution of Layers on the Web

From z-index clashes to native browser solutions

1998

The



# Thank you! 🙏

Let's stay in touch!

